정성훈 (Seonghoon Jeong)

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About.—I investigate and address system and network security challenges through a data-driven approach grounded in machine learning and deep learning. My work draws on a wide range of real-world data, including server-side online game transactions, root DNS server traffic, mobile payment records, and in-vehicle network intrusion datasets. I specialize in identifying and explaining intrusions in connected vehicles, notably with anomaly detection models trained solely on benign data.

My current research centers on trustworthy network intrusion detection built on foundation models, moving beyond classification accuracy toward systems that are robust, adaptive, and explainable. Key directions include multi-modal traffic representation learning that integrates payload bytes, packet sequences, and protocol metadata; drift-resilient detection that combines unsupervised drift metrics with parameter-efficient adaptation; and generative and causal reasoning that explains the cause of a threat and suggests response scenarios. These directions are applied to in-vehicle networks and to the detection of algorithmically generated malicious domains, and are being extended to system-level telemetry from Linux, Windows, and macOS. Together, this effort is directed toward the core technologies for a Foundation Model for Cybersecurity—a large-scale AI capable of identifying, explaining, and responding to threats across complex digital infrastructures. This vision is grounded in my expertise in both offensive and defensive security, bridging advanced AI methodologies with a deep, fundamental understanding of the systems being protected.

I have academic experience in teaching practical system hacking and defense techniques—such as reverse engineering, network hacking, smart car hacking, and attack–defense strategies (*e.g.*, machine learning-based intrusion detection systems)—as well as core courses including computer programming, algorithms, system security, and server administration. I believe such expertise is achieved through an in-depth understanding of traditional computer science and engineering concepts—including data structures, operating system concepts, system programming, data communications, and networking—so as to **really** understand security risks, threats, vulnerabilities, and proper countermeasures.

EDUCATION

Korea University, Seoul, Korea Mar 2017–Feb 2023
Ph.D. in Information Security
- Dissertation: Network Intrusion Detection and Prevention System for Connected Vehicles
- Advisor: Prof. Huy Kang Kim
- Focus: Attack model study against IEEE 1722 AVB applications and Automotive Grade Linux, and then developing an intrusion detection and prevention system

Korea University, Seoul, Korea Mar 2015–Feb 2017
M.S. in Information Security
- Thesis: A Longitudinal Analysis of .i2p Leakage in the Public DNS Infrastructure
- Advisor: Prof. Huy Kang Kim
- Focus: Measurement of DNS privacy leakage from Invisible Internet Project (I2P)

Chungbuk National University, Cheongju-si, Chungbuk, Korea Mar 2011–Feb 2015
B.S. in Information and Communication Engineering

- Capstone Project: Design and Implementation of a Forensics tool designed for Windows 7 and Windows 8
- Advisor: Prof. Min Choi
- Focus: Recovering Deleted Files from Windows NT File System

EMPLOYMENT

Division of Artificial Intelligence Engineering, Sookmyung Women's University Sep 2024–Present
Job title: Assistant Professor (Director, System and Network Security Lab.)

BK21 FOUR R&E Center for Cybersecurity, Korea University Mar 2024–Aug 2024
Job title: Postdoctoral Research Associate

Korea University Mar 2024–Aug 2024
Job title: Lecturer

Institute of Cyber Security & Privacy, Korea University Mar 2023–Feb 2024
Job title: Postdoctoral Research Associate

Korea University Mar 2023–Aug 2023
Job title: Lecturer

Korea University Mar 2022–Aug 2022
Job title: Lecturer

Institute of Cyber Security & Privacy, Korea University Mar 2018–Feb 2022
Job title: Technical Research Personnel (전문연구요원)

PUBLICATIONS

The following list is also available at Google Scholar (#of citations $\geq$ 1730, h-index 11, as of Oct 13, 2026) and 0000-0001-5638-2851. SciVal FWCI=7.24 (2020–2025) as of Aug 27, 2025.

JOURNAL (10)

[1] X-SPUR: Explainable Surprisal-Based Protocol-Aware Unsupervised Reasoning for Automotive Ethernet Intrusion Detection
J Kim, **S Jeong**
IEEE Transactions on Industrial Informatics (Early Access, Oct 2026) (SCIE), doi: 10.1109/TII.2026.3736030
Remark: **Corresponding author**; the first intrinsically explainable deep learning IDS for automotive Ethernet, led by an undergraduate student in my lab; rating Q1—JCR 2025 top 6.8% journal in category AUTOMATION & CONTROL SYSTEMS (6/88), 2025 JIF 9.8

[2] J1939DB-IDS: SAE J1939 Dual-Branch Intrusion Detection System Against Novel Attacks
H Lee, **S Jeong**, HK Kim
IEEE Transactions on Network and Service Management (2026) (SCIE), doi: 10.1109/TNSM.2026.3706666
Remark: Collaboration with Korea University; IDS for SAE J1939 heavy-duty vehicle networks against novel

attacks; rating Q1—JCR 2025 top 19.9% journal in category COMPUTER SCIENCE, INFORMATION SYSTEMS (53/266)

[3] AERO: Automotive Ethernet Real-Time Observer for Anomaly Detection in In-Vehicle Networks
S Jeong, HK Kim, ML Han, BI Kwak
IEEE Transactions on Industrial Informatics (Mar 2024) (SCIE), doi: 10.1109/TII.2023.3324949
Remark: The first unsupervised network-IDS for heterogeneous automotive Ethernet communications, rating Q1—JCR top 1.5% journal in category COMPUTER SCIENCE, INTERDISCIPLINARY APPLICATIONS (3/169) in 2023, **#citations** $\geq$ **45**

[4] X-CANIDS: Signal-Aware Explainable Intrusion Detection System for Controller Area Network-Based In-Vehicle Network
S Jeong, S Lee, H Lee, HK Kim
IEEE Transactions on Vehicular Technology (Mar 2024) (SCIE), doi: 10.1109/TVT.2023.3327275
Remark: The first unsupervised network-IDS for controller area network that identifies a specific target under attack, rating Q1—JCR top 12.2% journal in category TELECOMMUNICATIONS (15/119) in 2023, **#citations** $\geq$ **90**

[5] Detecting Domain Names Generated by DGAs with Low False Positives in Chinese Domain Names
H Lee, JD Yoo, **S Jeong**, HK Kim
IEEE Access (Sep 2024) (SCIE), doi: 10.1109/ACCESS.2024.3454242

[6] Trading Behind-the-Scene: Analysis of Online Gold Farming Network in Auction House System
Y Noh, **S Jeong**, HK Kim
IEEE Transactions on Games (Jul 2021) (SCIE), doi: 10.1109/TG.2021.3094054
Remark: Industrial collaboration (Netmarble corp.), fraud detection on a server-side online game service

[7] Convolutional Neural Network-based Intrusion Detection System for AVTP Streams in Automotive Ethernet-based Networks
S Jeong, B Jeon, B Chung, HK Kim
Vehicular Communications (Jun 2021) (SCIE), doi: 10.1016/j.vehcom.2021.100338
Remark: the first network-IDS for automotive Ethernet, Q1 rating—JCR top 8.065% journal in category TELECOMMUNICATIONS (8/93) in 2021, **#citations** $\geq$ **100**

[8] Cybersecurity for Autonomous Vehicles: Review of Attacks and Defense
K Kim, JS Kim, **S Jeong**, JH Park, HK Kim
Computers & Security (Apr 2021) (SCIE), doi: 10.1016/j.cose.2020.102150
Remark: **#citations** $\geq$ **570**

[9] Multimodal Game Bot Detection using User Behavioral Characteristics
AR Kang, **SH Jeong**, A Mohaisen, HK Kim
SpringerPlus (Apr 2016) (SCIE), doi: 10.1186/s40064-016-2122-8
Remark: **International & Industrial collaboration** (NCSOFT), **#citations** $\geq$ **55**, data-driven system security

[10] Analysis of Game Bot's Behavioral Characteristics in Social Interaction Networks of MMORPG
SH Jeong, AR Kang, HK Kim
ACM SIGCOMM Computer Communication Review (Aug 2015) (SCIE)—doi: 10.1145/2829988.2790005, ACM SIGCOMM (**SIGCOMM 2015**—a BK21-distinguished conference)
Remark: **One of the Top Conferences**, Acceptance rate: **17%**, **Citation counts** $\geq$ **29**, server-side data analysis

CONFERENCE/WORKSHOP (12)

[11] Cleansing Label Contamination in Popularity-based Ranking Lists for Robust DGA Detection
C Jung, C Lee, **S Jeong**
IEEE International Conference on Future Machine Learning and Data Science (**FMLDS 2026**), Kobe, Japan, Nov 2026 (accepted as a full paper, to appear)
Remark: Undergraduate research in my lab; shows that popularity-based ranking lists used as benign data contain morphologically anomalous domains and proposes a log-likelihood-ratio cleansing method

[12] NPP-ITA: Normal-Prototype Prior for Unsupervised Video Anomaly Detection with Inference-Time Adaptation
N Kim, M Kim, **S Jeong**
IEEE VTS Asia Pacific Wireless Communications Symposium (**APWCS 2026**), Seoul, Korea, Aug 2026
Remark: Undergraduate graduation research supervised by me

[13] DRIFT: Drift-Resilient Invariant-Feature Transformer for DGA Detection
C Lee*, C Jung*, **S Jeong**
IEEE/IFIP International Conference on Dependable Systems and Networks (**DSN 2026**—a BK21-distinguished conference), Charlotte, NC, USA, Jun 2026, doi: 10.1109/DSN69566.2026.00077
Remark: **Corresponding author**; **One of the Top Conferences** in dependable and secure computing, Acceptance rate: **19.6%** (65/331); *equal contribution by two undergraduate students in my lab, both awarded DSN'26 Student Travel Grants; source code and the companion dataset [28] are publicly released

[14] Lightweight Preprocessing Defenses for Robust Deepfake Detection Against Adversarial Perturbations
A Zhang, **S Jeong**, H Kang, DD Kim
The 5th ACM Workshop on the Security Implications of Deepfakes and Cheapfakes (**WDC 2026 Workshop**), co-located with ACM AsiaCCS 2026, Bangalore, India, Jun 2026, doi: 10.1145/3803629.3803673
Remark: **Corresponding author**; **International collaboration** (The University of Queensland, Australia)

[15] A Black-Box Inversion Attack on Small Language Models and Differential Privacy-Based Defense
X Wang, **S Jeong**, H Kang, DD Kim
ACM Workshop on Privacy in Large Language Models (LLM) and Natural Language Processing (NLP) (**LM-SHIELD 2026 Workshop**), co-located with ACM AsiaCCS 2026, Bangalore, India, Jun 2026, doi: 10.1145/3803628.3807971
Remark: **International collaboration** (The University of Queensland, Australia)

[16] The Vehicle May Be Sick: Denial of Diagnostic Services by Exploiting the CAN Transport Protocol
S Baek, **S Jeong**, HK Kim
Embedded Security in Cars Conference (**escar USA 2026**), arXiv:2604.23617
Remark: A premier automotive cybersecurity conference bridging industry and academia, Acceptance rate ≈**30%**; **unveiled a new kind of threat** that denies diagnostic services by exploiting the CAN transport protocol

[17] Infotainment System Matters: Understanding the Impact and Implications of In-Vehicle Infotainment System Hacking with Automotive Grade Linux
S Jeong, M Ryu, H Kang, HK Kim
ACM Conference on Data and Application Security and Privacy (**CODASPY 2023**), doi: 10.1145/3577923.3583650
Remark: **The Best Paper**, Acceptance rate ≈**30%**, Reported 3 vulnerabilities to the CVE database as well as 2 cases of vulnerability inheritances in a Linux OS. 한국정보과학회 우수학술대회, **Citation counts** ≥ **39**

[18] MUVIDS: False MAVLink Injection Attack Detection in Communication for Unmanned Vehicles
S Jeong, E Park, KU Seo, JD Yoo, HK Kim
International Workshop on Automotive and Autonomous Vehicle Security (**AutoSec 2021 Workshop**), co-

located with NDSS 2021, doi: 10.14722/autosec.2021.23036
Remark: Intrusion Detection System to secure **Unmanned Aerial System**, **Citation counts** $\geq$ **30**

[19] Automated Reverse Engineering and Attack for CAN using OBD-II
TU Kang, HM Song, **S Jeong**, HK Kim
IEEE Vehicular Technology Conference (**VTC-Fall 2018**), doi: 10.1109/VTCFall.2018.8690781
Remark: **Unveiled a new kind of threats** leveraging the self-diagnostics system, **Citation counts** $\geq$ **65**

[20] OTIDS: A Novel Intrusion Detection System for In-vehicle Network by using Remote Frame
H Lee, **SH Jeong**, HK Kim
Conference on Privacy, Security and Trust (**PST 2017**), doi: 10.1109/PST.2017.00017
Remark: **Citation counts** $\geq$ **610**

[21] Transparency in the New gTLD Era: Evaluating the DNS Centralized Zone Data Service
AR Kang, **SH Jeong**, SY Ko, K Ren, A Mohaisen
IEEE Workshop on Hot Topics in Web Systems and Technologies (**HotWeb 2016 Workshop**), doi: 10.1109/HotWeb.2016.18
Remark: **International collaboration**

[22] A Longitudinal Analysis of .i2p Leakage in the Public DNS Infrastructure
SH Jeong, AR Kang, J Kim, HK Kim, A Mohaisen
ACM SIGCOMM (**SIGCOMM 2016**—a BK21-distinguished conference), doi: 10.1145/2934872.2960423, Poster
Remark: **One of Top Conferences**, **International collaboration**, Acceptance rate: **17%**, **Citation counts** $\geq$ **12**, studies on 2 root DNS servers—A and J maintained by Verisign Inc.

DOMESTIC JOURNAL (4)

[23] 가상 개발환경 기반의 차량용 사이버훈련 프레임워크 설계: 공격 중심으로 (Design of a Virtual Development Environment-based Cyber Training Framework for Vehicles: Focused on Attacks)
조영복, 최수빈, 오병윤, 김호준, 최영호, **정성훈**, 곽병일, 한미란
정보보호학회지 (Review of KIISC) 33(4), 23–29 (Aug 2023)

[24] 차량 네트워크 공격 대응을 위한 가상화 사이버 훈련 개발환경 사례 연구 (A Case Study on a Virtualized Cyber Training Development Environment for Responding to Vehicle Network Attacks)
김호준, 최영호, 조영복, 최수빈, 오병윤, **정성훈**, 곽병일, 한미란
정보보호학회지 (Review of KIISC) 33(4), 31–40 (Aug 2023)

[25] The Big Data Analysis Framework of Information Security Policy based on Security Incidents
SH Jeong, HK Kim, J Woo
한국컴퓨터정보학회논문지 (Oct 2017), KCI, doi: 10.9708/jksci.2017.22.10.073

[26] A Survey of Fraud Detection Research based on Transaction Analysis and Data Mining Technique
SH Jeong, H Kim, S Shin, T Lee, HK Kim
정보보호학회논문지 (Dec 2015), KCI, doi: 10.13089/JKIISC.2015.25.6.1525

DOMESTIC CONFERENCE (1)

[27] Toward Understanding Network Intrusion on Vehicles: An eXplainable IDS Analyzing Signals and Time-series Periodicity
S Jeong
Conference on Information Security and Cryptography-Winter 2023 (CISC-W'23)
Remark: **Outstanding Paper Award** (정보보호학회장상-우수논문상)

DATASETS (4)

[28] Longitudinal Benign and DGA Domain Name Dataset
C Lee, C Jung, **S Jeong**
IEEE DataPort (2026), doi: 10.21227/za2s-9e09
Remark: Companion dataset of DRIFT [13]

[29] X-CANIDS Dataset (In-Vehicle Signal Dataset)
S Jeong, H Kang, HK Kim
IEEE DataPort (2024), doi: 10.21227/epsj-y384

[30] Automotive Ethernet Intrusion Dataset
S Jeong, B Jeon, B Chung, HK Kim
IEEE DataPort (2021), doi: 10.21227/1yr3-q009

[31] CAN Dataset for Intrusion Detection (OTIDS)
H Lee, **SH Jeong**, HK Kim
HCRLab, Korea University (2018), ocslab.hksecurity.net; IEEE DataPort, doi: 10.21227/sy7h-9057

PATENTS

Summary. 11 distinct patent items—6 registered in Korea (one more allowed in 2026) and 5 registered in the US, with further applications pending in Korea, Germany, and China.

Apparatus and Method for Detecting Vehicle Intrusion
- Korea, **Reg. No. 10-1853676**, Apr 2018

Device and Method for Analysing CAN Message Using OBD-II Query
- Korea, **Reg. No. 10-2028653**, Sep 2019

Device for Verifying Status and Detecting Anomaly of Vehicle and System Having the Same
- PCT/KR2018/009508, Aug 2018
- Korea, **Reg. No. 10-1995903**, Sep 2019

Method for SDN-based Intrusion Response or Prevention for In-vehicle Network and System using the same
- PCT/KR2020/010141, Jul 2020
- Korea, App. No. 10-2020-0095518, Jul 2020 (**Decision to grant**, 2026)
- US, App. No. 17/631,836, Jan 2022

Method for SDN-based Intrusion Response or Prevention for In-vehicle Network and System using the same
- Korea, **Reg. No. 10-2870304**, Oct 2025
- US, **Reg. No. US12127004B2**, Oct 2024
- PCT/KR2020/010142, Jul 2020
- Germany, App. No. 112020003658.8, Jan 2022
- China, App. No. 202080055847.3, Jan 2022

Anomaly Detection Model Using Message ID Sequence on Unmanned Moving Objects
- Korea, **Reg. No. 10-2476359**, Dec 2022
- US, **Reg. No. US12444241B2**, Oct 2025

Generative Adversarial Network Model and Training Method to Generate Message ID Sequence on Unmanned Moving Objects

- Korea, **Reg. No. 10-2691125**, Jul 2024
- US, **Reg. No. US12608603B2**, Apr 2026

Lightweight Real-Time Anomaly Detection Method Using CAN Message Analysis and Neural Network Model
- Korea, App. No. 10-2021-0174328, Dec 2021
- PCT/KR2022/017082, Nov 2022
- US, **Reg. No. US12665916B2**, Jun 2026

Method for Detecting Anomalies of Lidar Point Cloud Data and Related Device
- PCT/CN2021/114368, Aug 2021
- US, App. No. 18/584,834, Feb 2024

Intrusion Detection Method and Apparatus
- PCT/CN2021/114574, Aug 2021
- US, **Reg. No. US12585016B2**, Mar 2026 (App. No. 18/587,159, Feb 2024)

Model Pre-training Technology and Classification Method for Robust DGA-Generated Domain Detection
- Inventors: **S Jeong**, C Lee, C Jung
- Korea, App. No. 10-2026-0092483, 2026

AWARDS & SCHOLARSHIPS

Granite Tower Teaching Award—CYDF311 Class #37 Spring 2023, Spring 2024
- Top 5% of student course evaluation in Korea Univ.
- The President of Korea University

Outstanding Paper Award 2023
- Korea Institute of Information Security & Cryptology (한국정보보호학회 동계학술대회)

The Best Paper Award 2023
- Selected among 83 submitted papers
- ACM CODASPY 2023 Conference Organizing Committee

Excellent Teaching Award—AICS308 Reverse Engineering Spring 2022
- Top 20% of student course evaluation in Korea Univ.
- The President of Korea University

BrainKorea21 Four Fellowship Program 2021
- Top 5% of Korea Univ. graduate students
- Outstanding international activities in ITU-T as a delegate of Republic of Korea

Teaching Assistant 2018S, 2018F
- Administration office of School of Cybersecurity, Korea University

Student Travel Grants 2016
- ACM SIGCOMM 2016 Conference Organizing Committee

TEACHING

Instructor

NOTE: the number represents the score measured by students at the end of each semester. Full course reviews are available at www.snsec.net/teaching.

Basic Hacking Practice 2026F
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- Newly established major elective for 3rd-year students

21105590: System Security 2026F
- Division of Artificial Intelligence Engineering, Sookmyung Women's University

21102524: Introduction to Computer Programming 2026S
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- 4.93/5.00 in division 001 (exclusive section for AI Engineering freshmen of 2026)
- 4.73/5.00 in division 003

21105590: System Security 2025F
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- 4.92/5.00

21100720: Algorithms 2025F
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- 5.00/5.00

21102524: Introduction to Computer Programming 2025S
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- 4.99/5.00

21105541: Data-Driven Security (Graduate school) 2025S
- Dept. IT Engineering, Sookmyung Women's University
- 5.00/5.00

21102530: Server Administration and Security 2024F
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- 5.00/5.00 in division 001
- 5.00/5.00 in division 002

21100720: Algorithms 2024F
- Division of Artificial Intelligence Engineering, Sookmyung Women's University
- 4.95/5.00

CYDF311: Class #37—*Attack & Defense: Theory and practice* 2024S
- Dept. Cyber Defense, Korea University
- 4.95/5.00

CYDF311: Class #37—*Attack & Defense: Theory and practice* 2023S
- Dept. Cyber Defense, Korea University
- 5.87/6.00

AICS308: Reverse Engineering 2022S
- Dept. AI Cybersecurity, Sejong Campus of Korea University
- 5.67/6.00

Teaching Assistant

Lectured by Prof. Huy Kang Kim
- CYDF311: Class #37—*Attack & Defense: Theory and practice* 2021S
- IMS1004: Introduction Course for Information Security (Graduate School) 2021S

Lectured by Prof. Kyounggon Kim
- CYDF218: Class #66—*Cyber War* 2018F
- ISEC401: Advanced Hacking Practice 2018S
- ITCS302: Hacking and Security 2018S
- IMS756: Practical Cyber Security (Graduate school) 2018S
- CYDF218: Class #66—*Cyber War* 2017F
- ISEC302: Basic Hacking Practice 2017F
- ITCS302: Hacking and Security 2017S
- ISEC401: Advanced Hacking Practice 2017S

Lectured by Dr. Sanghyun Cho
- CYDF211: Class #18—*System Programming* 2018S

Special Lecture and Seminar

Hands-on-Practice: Penetration Test and Intrusion Detection System Development 2023
- Hosted by the University of Queensland in Australia (May 8–12, 2023)

Hands-on-Practice: Intrusion Detection System Development with Machine Learning 2022
- Hosted 20+ undergraduate students from the University of Western Australia

Smart Vehicle Security Training Course: Controller Area Network 2020
- Audiences included researchers from small/mid-size enterprises and cybersecurity research institutes
- hosted by the Korea Internet & Security Agency (KISA)

Data-Driven Security in Data Science Era 2018
- to graduate students of SungKyunKwan University
- hosted by SungKyunKwan University

KB-KAIST AI Intensive—A Practice on Detecting Fraudulent Credit Card Transactions 2018
- to data scientists in KB Kookmin Bank, Korea
- hosted by Korea Advanced Institute of Science and Technology (KAIST)

SERVICES

International Standard

Since Sep 2020, I have been attending ITU-T SG17 meetings as one of the delegates who represent the Republic of Korea (MSIT). I am the main editor of ITU-T X.1377 in ITU-T SG17 (Security)/WP2/Q13 (Intelligent Transportation Systems).
- **ITU-T X.1377** (formerly X.ipscv): Guidelines for an intrusion prevention system for connected vehicles, **Approved** in Oct 2022, see ITU-T work programme
- Adopted as a Korean National Standard, **KS X 3297** (Jan 2026)

Conference Organization and Program Committee

- **Program Chair**, Workshop on AI Dependability (**AID'25**), co-located with IEEE PRDC 2025 (Nov 4, 2025)
- **Publicity Co-chair**, the 31st IEEE Pacific Rim International Symposium on Dependable Computing (**PRDC 2026**)
- **Technical Program Committee**, IEEE International Conference on ICT Convergence (**ICTC 2026**)
- **Program Committee**, Conference on Information Security and Cryptography—Summer 2026 (CISC-S'26, Busan, May 7–8, 2026) and Winter 2025 (CISC-W'25), Korea Institute of Information Security & Cryptology
- **Program Committee** (paper reviewer), Korea Digital Forensics Society Summer Conference 2026 (Jun 25, 2026)
- **Session Chair**, the 1st KSAE Automotive Cybersecurity Research Group Workshop (한국자동차공학회 자동차 사이버보안연구회 워크숍), 2025
- **CSC 2021** Cyber Security Challenge 2021 "사이버 보안 챌린지 2021"
  –Host: Ministry of Science and ICT / Organizer: IITP / Committee: Korea Univ., Culture Makers, KISA
  – My role includes—Setting up an infotainment system testbed, Attack vector analysis, Penetration test, Vulnerability identification and disclosure, and the Development of a journal log collection framework for host-based IDSs.
  – I have evaluated the PoC exploit codes submitted by 102 researchers.
  – Live feed: The Final Round

Invited Talks

- Invited lecture, Korea Multimedia Society 2025 Fall Conference (한국멀티미디어학회 2025 추계학술대회), Nov 15, 2025
- Invited research presentation, "MetaFTT: Meta-Learning-Based FT-Transformer for Few-Shot IoT Intrusion Detection," Korea–Japan Database Workshop (KJDB 2025), Dec 12–13, 2025
- Presenter, Session 22 (Mobility Security), the 31st NetSec-KR 2025 (정보통신망 정보보호 컨퍼런스), COEX, Seoul, Apr 17–18, 2025

External Reviewer

- Elsevier Computers & Security
- Elsevier Internet of Things
- IEEE Transactions on Vehicular Technology
- IEEE Transactions on Industrial Cyber-Physical Systems
- IEEE Transactions on Neural Networks and Learning Systems
- IEEE Networking Letters
- SAE International Journal
- Springer Nature Scientific Reports
- IEEE International Conference on Future Machine Learning and Data Science (FMLDS 2026)
- Korea Multimedia Society MITA 2025

Maintainer of public datasets

- **CAN Dataset for intrusion detection** (**OTIDS**) available at HCRLab and IEEE DataPort (doi: 10.21227/sy7h-9057)
- **Automotive Ethernet Intrusion Dataset** available on IEEE DataPort (doi: 10.21227/1yr3-q009)
- **X-CANIDS Dataset** (**In-Vehicle Signal Dataset**) available on IEEE DataPort (doi: 10.21227/epsj-y384)
- **Longitudinal Benign and DGA Domain Name Dataset** available on IEEE DataPort (doi: 10.21227/za2s-9e09)

Vulnerability disclosure

- **CVE-2022-24595**
  Impact: Remote Code Execution, Information Disclosure, and Denial of Service
  `/usr/bin/afb-daemon` of Automotive Grade Linux 11.0.0–11.0.5 (Kooky Koi)
- **CVE-2022-24596**
  Impact: Denial of Service
  `/usr/bin/afb-daemon` of Automotive Grade Linux 11.0.0–11.0.5 (Kooky Koi)
- **CVE-2022-24597**
  Impact: Information Disclosure, Denial of Service
  `agl-service-weather` of Automotive Grade Linux 11.0.0–11.0.5 (Kooky Koi)

Technology transfer

- KU Research & Business Foundation → K-SIGN
  Patent No. 10-2021-0174328 Lightweight Real-Time Anomaly Detection Method Using CAN Message Analysis and Neural Network Model, 5,000,000 KRW

RESEARCH PROJECTS

Summary. As a graduate student researcher and a postdoctoral research associate, I have participated in 28 research projects funded by the Republic of Korea government as well as industries both internationally and at home.

Vehicle Cybersecurity

1. 자율주행차량의 차세대 내부 네트워크의 보안 및 초고속 무결성 부여 기술 개발, 정보통신기획평가원, Mar 2023–Feb 2024
2. 자동차 사이버보안 관리시스템 평가기술 및 보안위협 대응방안 기술 개발, 국토교통과학기술진흥원, Mar 2023–Feb 2024
3. 차세대보안 분야 챌린지 R&D 총괄 과제, 정보통신기획평가원, Apr 2020–Dec 2021, Apr 2023–Dec 2023
4. 가상화 기반 자율주행차 사이버훈련 환경 구축 및 콘텐츠 연구, 국가보안기술연구소, Apr 2023–Oct 2023
5. 무인이동체 공통핵심 보안기술 개발 연구, 정보통신기획평가원, Apr 2020–Sep 2022
6. Advanced Contextually-Aware On-board Monitoring for Self-Driving Cars, HUAWEI Singapore, Jul 2021–Apr 2022 (**International collaboration**)
7. Intrusion Resilient Sensor Perception for Autonomous Driving, HUAWEI Singapore, Aug 2020–Aug 2021 (**International collaboration**)
8. OTAC 인증 review task, 센스톤 주식회사, Aug 2021
9. IoT 관점에서의 차량 보안, 삼성전자주식회사, Nov 2015–Nov 2020 (**Long-term commitment**)
10. 차량 내부 네트워크 용 IDS(Intrusion detection System) 및 IRS(Intrusion Response System) 관련 국제 표준 개발, 현대NGV, Aug 2019–Feb 2020
11. 이더넷기반 차량 환경 공격발생시 자동 대응 방안 연구, 한국전자통신연구원, Jun 2019–Oct 2019
12. 자율주행 스마트자동차용 이상징후 탐지 핵심기술개발, 정보통신기술진흥센터, Jan 2017–Dec 2018
13. 다중 도메인 차량 이상징후 탐지 알고리즘 연구, 한국전자통신연구원, Aug 2018–Nov 2018
14. 차량 침입 탐지 기술, 삼성전자주식회사, May 2016–Dec 2016

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1. 4단계 BK21 스마트시티 정보보호학 교육연구단, 교육부&한국연구재단, Mar 2024–Aug 2024
2. System call 데이터의 filtering과 feature engineering을 통한 이상징후 탐지 패턴 생성, 주식회사 소테리아, Feb 2020–Jun 2020
3. AI기반 악성파일 탐지 모델 개발, 삼성전자주식회사, Apr 2019–Nov 2019

4. 사이버 공격 예측 및 자산 위험도 정량적 평가 기법 연구, 국방과학연구소, Mar 2018–Dec 2018
5. 스크립트 기반 사이버 공격 사전 예방 및 대응 기술 개발, 한국인터넷진흥원, Feb 2016–Feb 2017
6. 침입 탐지 방지 솔루션 연구, 삼성전자 주식회사, Jun 2015–Nov 2015

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1. 광고사기탐지 프로젝트, AISpera, Feb 2021–Feb 2022
2. A3 시계열 이상 탐지 고도화 프로젝트, 넷마블주식회사, Aug 2020–Nov 2020
3. 게임 내 소셜 네트워크 분석을 통한 게이머간 상호 작용 분석 한국전자통신연구원, Apr 2018–Nov 2018
4. 온라인게임 내 작업장 알고리즘 개발연구, 넷마블주식회사, Jan 2018–May 2018
5. 인게임 환경 요소 변화와 게이머 행동 변화간 상관관계 연구, 한국전자통신연구원, May 2017–Nov 2017
6. 모바일 온라인 RPG 게임 플레이 활성화 요소 연구, 한국전자통신연구원, Jun 2016–Nov 2016
7. 데이터마이닝 기법을 통한 게임 봇 탐지 및 악성유저 행위 분석: 지하경제 분석을 중심으로, 한국연구재단, Sep 2015–Apr 2017
8. 텍스트 마이닝 및 빅데이타 감성분석 기반 기업이미지 관리 시스템 구축, 한국연구재단, Sep 2015–May 2016
9. 모바일 결제 정보에 대한 익명화 및 이상거래 탐지 기술 연구, 한국인터넷진흥원, Jul 2015–Jan 2016

End of CV. Last update on October 13, 2026.